

PUBLIC / EXTERNAL

QynSight Guide

Quenyx vOPS HUB — v1.0.0 RC1 · Document v2.1

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REVISION HISTORY

Version	Date	Notes
1.0	2026	Initial v1 pack (through Sprint 19).
2.0	2026-06-29	Aligned to v1.0.0 RC1; native monitoring engine (no Nagios dependency).
2.1	2026-06-30	Added Operations Intelligence (Sprint 21): Monitoring Copilot, Alert/Root-Cause/Capacity/Performance/Infrastructure/Service-Health intelligence, evidence-based recommendations, and the Operations Intelligence dashboard — all reusing the Quenyx AI Platform.

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Audience: QynSight users and admins. **Status:** ● Production-ready. Core monitoring is **feature-frozen** at v1.0; **Operations Intelligence** (Sprint 21) is the active intelligence layer built on top of the frozen monitoring surface.

1. Overview

QynSight is Quenyx's operations/observability module ("observe"). It monitors hosts and services, maps infrastructure, analyzes performance, plans capacity, manages alerts, runs service checks, and enrolls agents — all scoped per workspace and gated by the `qynsight` module entitlement.

2. Real-time monitoring

Live metrics per host/service: `observe/real-time/metrics`, `real-time/system-info`, `real-time/thresholds`, plus `instances` and `instances/summary`. **Requires the Laravel scheduler** (cron running `observe:run-checks` each minute); without it data shows "never".

3. Infrastructure map

Topology and connections: `observe/infrastructure/topology` and `infrastructure/connections`. The UI renders the map and supports export (JSON/PNG).

4. Performance analytics

`observe/performance/metrics` and historical data (`observe_metrics_history` table) provide performance trends and threshold context.

5. Capacity planning

`observe/capacity-planning` (+ `/export`) and `observe/capacity/metrics` provide a capacity advisor and exportable capacity data.

6. Alert management

- **Rules:** list/create/update/delete + **toggle** (`observe/alerts/rules...`).
- **Events:** `alerts/history`, acknowledge via `alerts/events/{event}/acknowledge`.
- **Summary & channels:** `alerts/summary`, `alerts/channels`.
- **Monitoring profile:** `observe/monitoring-profile` (GET/PUT).

7. Service checks

`observe/services` , `observe/service-definitions` , and `observe/run-checks` — **native** service checks (HTTP, TCP, Ping, and custom plugin scripts) executed by the QynSight Monitoring Engine on a per-service check/retry interval. No external monitoring daemon is required. Reports:

`observe/reports` , `reports/summary` .

8. Hosts

Managed via **targets**: `observe/targets` (GET/PUT), `targets/validate` , and per-host port scan (`targets/{hostId}/port-scan`). Hosts carry address/public-IP metadata (`observe_targets_hosts`).

9. Agent enrollment

- Generate an **enrollment token** (`agents/enrollment-token`), list/revoke tokens, view metadata.
- Download the agent binary: `GET /api/agents/download/{platform}` (built on demand if Go is configured; see Doc 10).
- Agents authenticate with token/secret and push `register` , `heartbeat` , `metrics` , `inventory` .

10. Alert rules

Define conditions and channels; toggle rules on/off; review fired events and acknowledge them. Tied to monitoring profiles per workspace.

11. Capacity advisor

Derived from capacity metrics; exportable. Use it to anticipate resource constraints.

12. Exports

Infrastructure map (JSON/PNG) and capacity planning export are supported.

13. Port scans

On-demand and scheduled port scans run as **background jobs** — a **queue worker** is required (`infrastructure/port-scans` + `/run` , `targets/{hostId}/port-scan`). Results in `host_port_scans` tables.

14. Known limitations

- Requires scheduler + queue worker to be running for live data and scans.
- Agent on-demand build needs Go configured on the server (otherwise pre-place binaries).

- v1.0 is **feature-frozen** — no functional changes pre-Sprint 20.

15. Feature-freeze status

QynSight's **core monitoring engine** (checks, metrics, alerts, topology, capacity, agents) is **frozen** at v1.0. Section 16 (Operations Intelligence) is an additive intelligence layer that reads the frozen monitoring data — it does not change monitoring behavior.

16. Operations Intelligence (Sprint 21)

Operations Intelligence turns QynSight from a monitoring platform into an **operations intelligence platform**: the AI *understands and explains* your operational data rather than just answering free questions. It **reuses the Quenyx AI Platform** (Doc 07) end-to-end — the same provider abstraction, prompt orchestration, conversation service, and audit — with **no duplicated AI logic**. Every narrative is grounded in **real** monitoring evidence; when evidence is insufficient the system says so and never fabricates data.

Access & scope. All Operations Intelligence endpoints are workspace-scoped, require the `qynsight` module entitlement, monitoring RBAC, and the per-workspace `can_use_ai` AI capability. Every AI request is audited, provider-logged, conversation-logged, and rate limited. All resource identifiers are **UUIDs** (a deterministic UUIDv5 is derived from internal monitoring IDs — no schema change and no numeric IDs are exposed).

16.1 Monitoring Copilot

Ask grounded operational questions — "Which hosts are unhealthy?", "Summarize today's alerts", "What changed in the last 24 hours?", "Which hosts will run out of storage first?". Answers use current hosts, services, alerts, capacity, metrics, and topology. Each thread is a **real Quenyx AI conversation** that can be opened in the AI Workspace.

16.2 Alert Intelligence — ✨ Explain / ✨ Investigate

Every alert gains **Explain** and **Investigate** actions that return: operational impact, most likely causes, the **evidence used**, related alerts, suggested actions, and a **confidence score only when derived from real evidence**.

16.3 Root Cause Analysis (deterministic)

A deterministic analyzer scores resource layers (CPU → memory → storage → database → application) from real alert history, metrics, dependencies, and topology, then identifies the most probable root cause and **explains why**. Causal chains are never invented.

16.4 Incident Timeline

Auto-generated timelines built from **actual event timestamps** (alert lifecycle and service state changes), e.g. alert opened → AI investigation → recovery.

16.5 Capacity Intelligence

Reuses Capacity Planning and adds per-host forecasting with an AI explanation of growth trend, forecast, estimated exhaustion, operational impact, recommended action, and business risk — from available historical metrics only.

16.6 Performance Intelligence

Explains performance degradation, trend changes, resource hotspots, slow services, infrastructure bottlenecks, and anomalies, with recommendations, from real metric history.

16.7 Infrastructure Intelligence

Reuses the Infrastructure Map topology to answer "which systems depend on this host?", blast radius if a host fails, critical paths, downstream services, single points of failure, and potential cascading failures.

16.8 Service Health Intelligence

Beyond Healthy/Warning/Critical, the AI explains **why** a service is in its state, what changed, the expected impact, suggested action, and related systems.

16.9 Operational Recommendations

Evidence-based recommendations (e.g. increase RAM, restart service, investigate database, check storage, review thresholds). Every recommendation references real metrics, alerts, capacity, or dependencies — never produced without evidence.

16.10 Contextual ✨ actions & dashboard

Existing QynSight pages get a single, uncluttered ✨ **Quenyx AI** action: Host → Explain, Service → Analyze, Alert → Investigate, Capacity → Predict, Infrastructure Map → Impact Analysis. A new **Operations Intelligence dashboard** (`/observe/operations-intelligence`) shows only real data: infrastructure health summary, open alerts, critical services, top operational risks, predicted capacity risks, recent recommendations, and recent AI investigations.

See Doc 08 (API Reference) §Operations Intelligence for the full endpoint contract and Doc 07 (AI Platform Bible) for how these features reuse the shared AI runtime.